

Mohammad Yousuf Qureshi

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EDUCATION

Georgia Institute of Technology — M.S. Analytics <i>Computational Data Analytics Track, CGPA: 3.83 / 4.0</i>	May 2026
Texas A&M University at Qatar — B.S. Mechanical Engineering <i>Magna Cum Laude, CGPA 3.74 / 4.0 Tau Beta Pi Pi Tau Sigma Dean's Honor Roll</i>	May 2020

ANALYTICS PROJECTS

Social Media Signals for Flight Booking Surges — Lufthansa Group <i>NDA Industry Project github.com/YousufQ9/reddit-flight-demand-signals</i> - Led social media analysis pipeline on ~45M rows of proprietary booking data; processed unstructured data from 5 travel subreddits using VADER sentiment, spaCy NER, and RapidFuzz fuzzy matching to map 221 destinations to ISO codes - Discovered ~60% of raw Reddit-booking correlation was spurious seasonality; applied STL deseasonalization to isolate genuine 7–28-day leading signals; Granger causality confirmed in 7 of 41 markets; final relevance score achieved AUC = 0.842 on high-signal markets and 9× lift in booking spike prediction	Spring 2026
Clinical Chest X-Ray Report Generation — LSTM vs. Transformer <i>github.com/YousufQ9/dl-cxr-report-gen</i> - Implemented hierarchical LSTM baseline (Jing et al. 2018) with DenseNet-121 encoder achieving BLEU-1 of 0.286 (+33% over default); built full evaluation module covering linguistic (BLEU, METEOR, CIDEr, ROUGE-L) and clinical metrics (CheXbert F1, hallucination rate) - Clinically guided Transformer with weighted cross-entropy loss improved CheXbert macro-F1 by 23% and CIDEr by 43% over vanilla Transformer, confirming a measurable fluency-clinical accuracy trade-off	Spring 2026
Inflationary Impact of U.S. Tariffs — Interactive Dashboard <i>github.com/YousufQ9/tariff_impact_dashboard</i> Built ARIMAX models (with tariff rates as exogenous variable) for electronics, food & apparel CPI forecasting using 10 years of BLS/FRED/USITC data; deployed results to an interactive Tableau CPI Simulator with 130+ public views, rated 4.56/5 by users	Fall 2025
Dog Sanctuary Management System — Database Application <i>github.com/YousufQ9/database_stevies_dog_sanctuary</i> - Designed and built a full-stack web application (React frontend, Django backend, MySQL database) for a fictional dog rescue organization, covering the complete lifecycle from requirements analysis to ER modeling, relational schema design, and implementation - Implemented role-based access control (Volunteer vs. Executive Director permissions) with session-based validation at the application and query level, and modeled complex business rules directly into the schema, including tiered adoption fee calculations and duplicate-expense prevention constraints	Spring 2025

EXPERIENCE

Estimation Engineer — Fire Protection Systems, JBK Controls, Qatar <i>Sprinkler, Deluge, Pre-Action, Foam, and Clean Agent (FM200 & NOVEC) Systems</i> - Forecasted material and labor costs for fire suppression projects using Excel-based cost models, referencing historical project data and supplier pricing to inform estimates - Extracted and structured quantity take-off data from AutoCAD drawings into detailed Excel cost templates; managed supplier negotiations and site assessments across sprinkler, foam, and clean-agent (FM200/NOVEC) systems - Analyzed and compared multi-vendor supplier quotations to identify cost-optimal procurement options during negotiations, balancing pricing against technical compliance requirements	Sep 2020 – May 2024
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TECHNICAL SKILLS

Languages & Tools: Python, R, SQL, Tableau, Power BI, Git, Jupyter, Docker, Excel, Data Visualization, Streamlit
Cloud & Big Data: PySpark, Apache Spark (DataFrame API), Databricks, AWS (S3, Athena), Google Cloud Platform
ML & Analytics: Scikit-learn, Statsmodels, ARIMAX, XGBoost, Random Forest, Clustering, Regression, Bayesian Inference, Matplotlib, Seaborn
Data Engineering: Pandas, NumPy, MySQL, Feature Engineering, Data Cleaning & Imputation, ETL Pipelines, Data Modeling, Schema Design
NLP: VADER, spaCy, RapidFuzz, NER, Text Preprocessing

CERTIFICATIONS

- Microsoft Azure Essentials — Professional Certificate (*Microsoft & LinkedIn Learning, Jul 2026*)
- Career Essentials in GitHub — Professional Certificate (*GitHub & LinkedIn Learning, Jul 2026*)
- Career Essentials in Generative AI (*Microsoft & LinkedIn Learning, Jul 2026*)
- Data Analytics Job Simulation (*Deloitte, via Forage, Jul 2026*)